

MIT ADT University, Pune
School of Computing
Department of Computer Science & Engineering
S.Y. B.Tech. Sem-3
23CSE1007-Processor Architecture and Interfacing



ASSIGNMENT No.4

Handed on :- 04st Oct 2025

Last date :- 15th Oct. 2025

1. Draw and label the pin diagram of the ATmega328 microcontroller and explain the function of any five pins.
 2. List and explain the basic features of the ATmega328 8-bit microcontroller.
 3. Explain the block diagram of a generic Arduino board and describe the function of each block.
 4. Compare Microcontroller with microprocessor.
 5. How does the Arduino UNO board utilize the ATmega328 microcontroller, and what additional features does the board provide for ease of use
 6. What is an Arduino board, and how is it useful for embedded system development and prototyping?
 7. What are the key differences between a microprocessor and a microcontroller in terms of architecture and application?
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Multiple Choice Questions

1. What is the main advantage of a microcontroller over a microprocessor in embedded systems?
A) Higher cost
B) Requires external components
C) Compact and self-contained
D) Slower processing speed
2. Which of the following components is typically not included within a microcontroller?
A) CPU
B) RAM
C) Hard Disk
D) I/O ports
3. Arduino UNO is built around which microcontroller?
A) ATmega16
B) ATmega328P
C) AT89C51
D) PIC16F877A
4. What is the operating voltage of an Arduino UNO board?
A) 9V
B) 12V
C) 3.3V
D) 5V

5. What is the operating voltage of an Arduino UNO board?
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 - B) 12V
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6. In Arduino, the USB interface is used for:
 - A) Power supply only
 - B) Internet access
 - C) Programming and communication
 - D) Connecting external storage
7. Which communication interfaces are available on Arduino UNO?
 - A) SPI
 - B) I2C
 - C) UART
 - D) All of the above
8. The ATmega328P has how many digital I/O pins?
 - A) 10
 - B) 14
 - C) 20
 - D) 32
9. What type of microcontroller is ATmega328?
 - A) 16-bit
 - B) 32-bit
 - C) 8-bit
 - D) 64-bit
10. What does AVR stand for in AVR microcontrollers?
 - A) Analog Voltage Register
 - B) Advanced Virtual RISC
 - C) Automatic Voltage Resistor
 - D) Arithmetic Virtual Register
11. How much Flash memory is available in ATmega328?
 - A) 8 KB
 - B) 16 KB
 - C) 32 KB
 - D) 64 KB

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